MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST -1 (SET-2)

Course: B.Tech. **Semester: 5th**
Session: 2024-25 **Section: A, B, C, D, I, J**
Subject Name: Database Management System **Subject Code: (BCS 501)**
Max Marks: 20 **Time: 1 hr**

Q .No.	1	2	3	4	5	6
CO No	CO1	CO2	CO2	CO2	CO1	CO1
Bloom's level	K4	K5	K2	K1	K3	K1

CO1: Understand Relational Data Model, Network Data Model and Hierarchical Data Model and Design new relational schema using Entity relationship diagrams to create normalized relations to manage information of company database.
CO2: Construct queries in relational algebra, tuple, domain calculus and SQL construct, and by applying integrity, key and referential integrity key constraints to manage information of company database.

Note:
1) This paper contains two sections (A) & (B).
2) All sections are compulsory.
3) Draw a neat diagram wherever required.

SECTION-A (06 Marks)

Q1. Distinguish between physical data independence and logical data independence with the help of example. **02 Marks**

Q2. Define the difference between left outer join and right outer join with suitable example. **03 Marks**

Q3. Consider the employee database. **03 Marks**
Employee (employee-name, street, city)
Works (employee-name, company-name, salary)
Company (company-name, city)
Manages (employee-name, manager-name)
Give an expression in relational algebra for each of the following queries.
a. Find the names of all employees who work for First Bank Corporation.
b. Find the names and cities of residence of all employees who work for First Bank Corporation.

Write the following queries Relational Algebra

SECTION-B (09 Marks)

Q4. Define the different Integrity Constraints? Explain with the help of suitable examples. **04 Marks**

Q5. Construct an E-R diagram for a hospital with a set of medical doctors and a set of patients. Associate with each patient a log of the various tests and examinations conducted.. **04 Marks**

Q6. Define the following terms with example: **04 Marks**
a) Weak entity set and Strong entity set
b) Foreign Key and discuss the concept behind declaration of foreign keys.

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SESSIONAL TEST -2 (SET-2)

Course: B.Tech. **Semester: 5th**
Session: 2024-25 **Section: A, B, C, D, I, J**
Subject Name: Database Management System **Subject Code: (BCS 501)**
Max Marks: 20 **Time: 1 hr**

Q .No.	1	2	3	4	5	6
CO No	CO3	CO4	CO5	CO3	CO4	CO5
Bloom's level	K2	K1,K2	K1	K1,K3	K3, K4	K2

CO3: Design company database's schema to solve the problem of null values, redundancy and anomaly by applying process of normalization.
CO4: Understand the dirty read, incorrect summary and lost update problem in transactions and examine serializability and recoverability of schedule for concurrent execution of transactions and apply in central and distributed database environment
CO5: Understand 2-phase Locking, Time Stamp Ordering, Validation based and multi-versioPn protocol to control concurrency.

Note:
1) This paper contains two sections (A) & (B).
2) All sections are compulsory.
3) Draw a neat diagram wherever required.

SECTION-A (06 Marks)

Q1. Discuss database anomaly with example? **02 Marks**

Q2. List ACID properties of transaction and usefulness of each. Explain the importance of log? **03 Marks**

Q3. Define lock in concurrency control. Discuss 2-PL concurrency control protocol. **03 Marks**

SECTION-B (09 Marks)

Q4. Define the normalization? Consider a relation schema R= (A, B, C, D, E) decompose into R1 (A, B, C) and R2 (A, D, E). Determine the Decomposition is a loss-less-join decomposition if the following set F of FD A→B C, CD→E, B→D, E→A **04 Marks**

Q5. Contrast between serial schedule and non serial schedule? Consider a schedule S for transactions T1 and T2 with set of read and write operations
S: R1(X); R2(X); R2(Y); W2(Y); R1(Y); W1(X).Determine that given schedule is Equivalent to serial schedule or not. **04 Marks**

Q6. Discuss time stamping protocol for concurrency control ? Also discuss Multiple Granularity. **04 Marks**